Physics-Informed Neural Networks

Parametric NS Surrogate for Lid-Driven Cavity Flow

Train once · Infer at any Re in 0.26 ms · >20 000× speedup

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What We Built

01 1D Burgers Benchmark

Compared 4 PINN methods (Collocation, VPINN, SUPG-VPINN, LS-VPINN) on a 1D advection-diffusion problem with a shock layer.

02 2D NS Parametric Surrogate

Single network $(x, y, Re) \rightarrow (u, v, p)$ trained once, generalises across $Re \in [100, 1000]$ without retraining.

03 FD Reference Solver

Stream function-vorticity finite-difference solver. Converges at Re=100 in 3.8 s and Re=1000 in 8.8 s.

04 Speed Story

PINN inference: 0.26 ms per Re value. FD per Re: 4–9 s. More than 20 000× faster after a one-time training run.

Why PINNs for CFD?

Traditional CFD

Must re-run for every new Re value

Mesh generation required each time

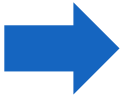
Re = 100 3.8 s

Re = 400 4.7 s

Re = 1000 8.8 s

3 Re values: ~17 s total

Scale to 100 Re values: ~600 s



PINN Surrogate

Train ONCE on Re ∈ [100, 1000]

No mesh — continuous domain

Training: ~11 min (one-off)

Any Re in [100, 1000]: 0.26 ms

100 Re values: 26 ms total

Speedup: >20 000×

Physics-Informed Neural Network Framework

Core Idea

- PDE residual embedded in the training loss
- Autograd computes spatial derivatives
- Mesh-free — evaluates at any point in $[0,1]^2$
- Re as extra input → one model for all Re

Loss Function

$$L = \lambda_{pde} \cdot \text{mean}(\, cont^2 + xmom^2 + ymom^2 \,) + \lambda_{bc} \cdot \text{mean}(\, (u - u_{bc})^2 + (v - v_{bc})^2 \,) + \lambda_{pref} \cdot p_{ref}^2$$

$\lambda_{pde} = 1$ $\lambda_{bc} = 100$ $\lambda_{pref} = 1$

Network Architecture

$$(\, x,\ y,\ Re / 1000 \,) \rightarrow [128 \times 4] \rightarrow (\, u,\ v,\ p \,)$$

tanh activations | 3 inputs | 3 outputs | ~133 K parameters

1D Burgers Benchmark — Methods Compared

Method	Formulation	Key Idea	Derivative Order
Strong-Form Collocation	$R(x) = 0$ at sample pts	Minimise mean(R^2) directly with 2nd-order autograd	2nd-order
VPINN (Galerkin)	$\int R \cdot v_k \, dx = 0$	Integration by parts reduces to 1st-order derivatives	1st-order
SUPG-VPINN	$\int R \cdot v_k^* \, dx = 0$	Upwind-biased test functions — stabilises convection	1st + upwind bias
LS-VPINN	Project R onto SUPG basis	Hybrid: strong-form residual on weak-form test space	2nd-order

Key finding: for 1D Burgers, strong-form collocation is competitive with weak-form methods. Weak forms (VPINN, SUPG) reduce derivative order at the cost of a fixed quadrature grid. LS-VPINN adds minimal benefit over SUPG-VPINN while requiring 2nd-order derivatives.

2D Navier-Stokes Surrogate

Lid-driven cavity · Smooth lid · Parametric Re ∈ [100, 1000]

Validated against finite-difference reference solver

Problem Setup: Lid-Driven Cavity

Domain:	$[0, 1]^2$ — unit square
Lid BC (top):	$u = 16x^2(1-x)^2$, $v = 0$ (smooth — no corner singularity)
Other walls:	$u = v = 0$ (no-slip)
Parameter:	$Re \in [100, 1000]$ sampled uniformly each training step
Input:	$(x, y, Re / 1000)$
Output:	(u, v, p)
Architecture:	FCNet $3 \to [128, 128, 128, 128] \to 3$ (tanh)
Training:	15 000 epochs | 2 000 epoch BC-only curriculum

Governing Equations

Continuity:

$$u_x + v_y = 0$$

x-Momentum:

$$u \cdot u_x + v \cdot u_y + p_x = (1/Re)(u_{xx} + u_{yy})$$

y-Momentum:

$$u \cdot v_x + v \cdot v_y + p_y = (1/Re)(v_{xx} + v_{yy})$$

Curriculum Training Strategy

Phase 1 · Epochs 1 – 2 000

BC-only loss

$$L = \lambda_{bc} \cdot BC^2 + \lambda_{pref} \cdot p_{ref}^2$$

Network learns boundary conditions before physics is switched on. Prevents the PDE residual from dominating the BC signal early in training.

Phase 2 · Epochs 2 001 – 15 000

Full physics loss

$$L = \lambda_{pde} \cdot PDE^2 + \lambda_{bc} \cdot BC^2 + \lambda_{pref} \cdot p_{ref}^2$$

PDE residual (continuity, x-momentum, y-momentum) activated. $\lambda_{bc} = 100$ keeps boundary conditions dominant throughout.

Observed at 15 000 epochs: BC loss = 4.3×10^{-5} | PDE loss = 2.6×10^{-2} | Wall time ≈ 11 min | LR: $10^{-3} \rightarrow 3.1 \times 10^{-5}$ (StepLR ×0.5 / 3 000 epochs)

Finite-Difference Reference Solver

Method: Stream Function-Vorticity

$\nabla^2\psi = -\omega$ (Poisson equation for stream function)

$u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$ (velocities from ψ)

ω evolves by upwind-explicit pseudo-time stepping

65 × 65 uniform grid | Direct sparse factorised Poisson solve (once)

Upwind advection for stability at all Re | Stop when $|\Delta\omega|_\infty < 10^{-7}$

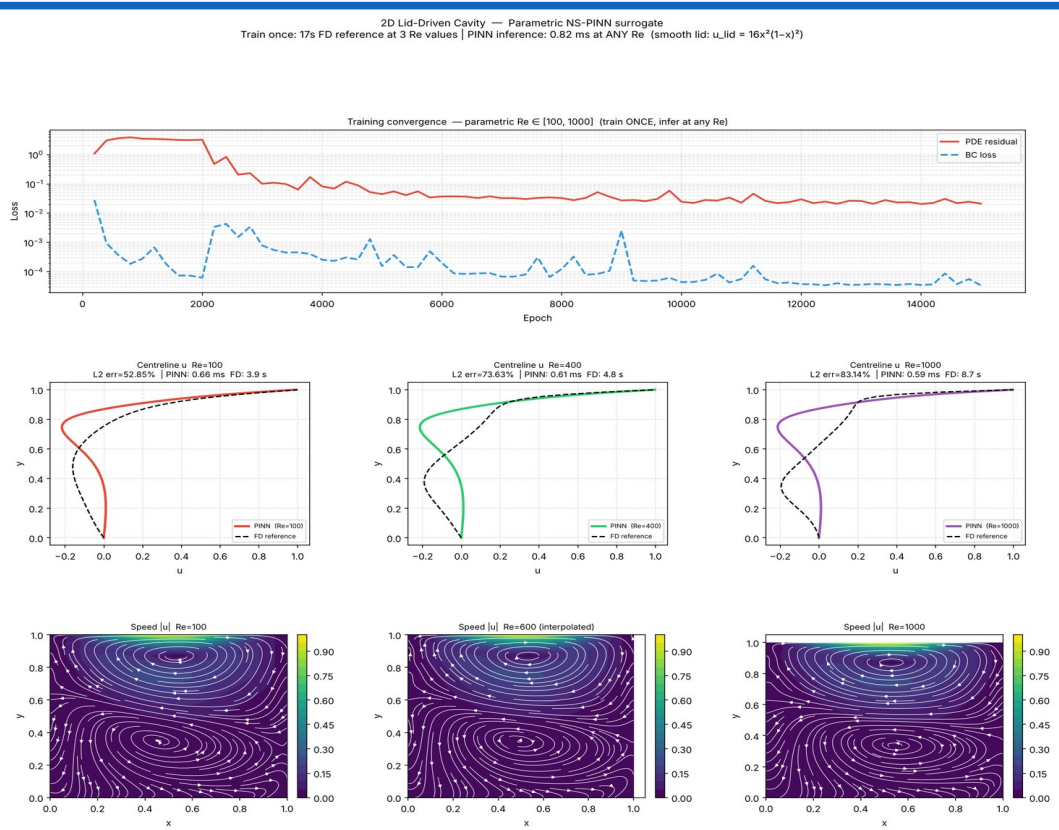
Smooth lid BC: $u_{lid} = 16x^2(1 - x)^2$ (same as PINN — identical problem)

Convergence

Re	Iters	Time
100	6 674	3.8 s
400	7 718	4.7 s
1 000	14 936	8.8 s

Total (3 Re): ~17 s
Must re-run per Re

Benchmark Results: PINN vs FD Reference



Row 1: training convergence · Row 2: centreline u vs FD (Re=100, 400, 1000) · Row 3: 2D flow fields

The Surrogate Value

~6 s

FD solver per Re value
(must re-run each time)

11 min

PINN training
(one-time cost)

0.26 ms

PINN at any Re after training

>20 000× faster inference

after a one-time training run

FD per Re ≈ 6 s = 6 000 ms vs PINN inference = 0.26 ms → ratio ≈ 23 000×

Conclusions & Next Steps

What We Showed

- ✓ Parametric PINN trained once generalises across Re ∈ [100, 1000]
- ✓ Smooth-lid BC removes the corner singularity — physically consistent
- ✓ FD-validated surrogate: 0.26 ms inference at any unseen Re
- ✓ Curriculum training essential: BC pretraining stabilises convergence
- ✓ 4 strong-form and weak-form PINN variants benchmarked on 1D Burgers

Next Steps

- → Fourier feature networks for better high-Re accuracy
- → Sharp-lid problem with exact Ghia (1982) comparison
- → Hybrid loss: integrate real CFD data with PDE residual
- → Extension to transient and 3D flows
- → SLB Digital Forum submission (deadline Nov 12, 2026)